

有机化学B专题讲座2

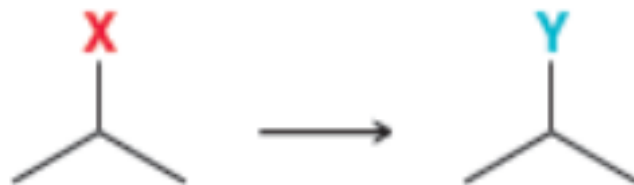
合成策略2

佟 硕

2025年6月7日

- One-Step Syntheses

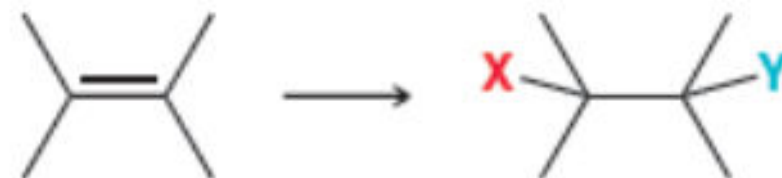
取代反应



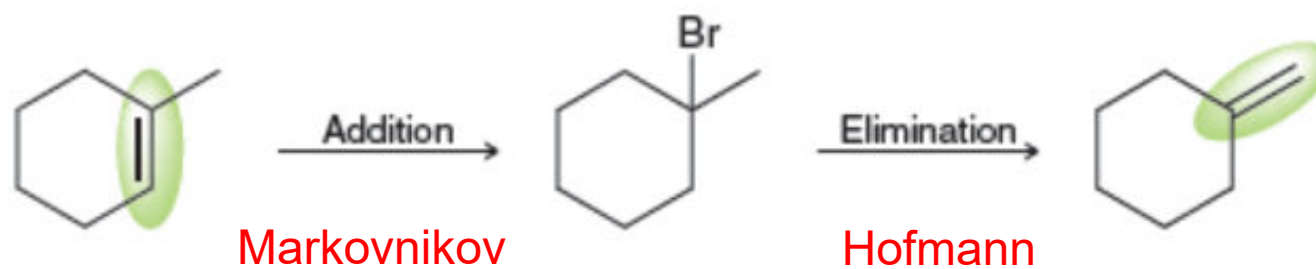
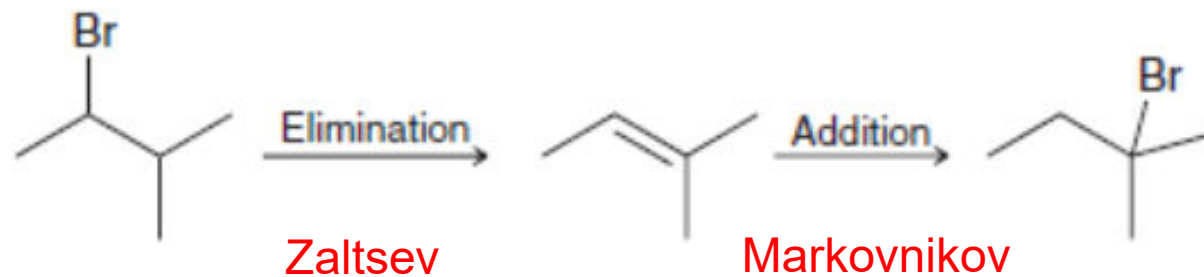
消除反应



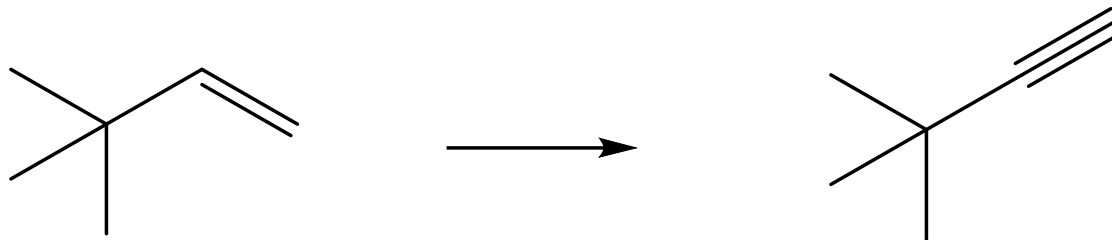
加成反应



- 改变官能团的位置



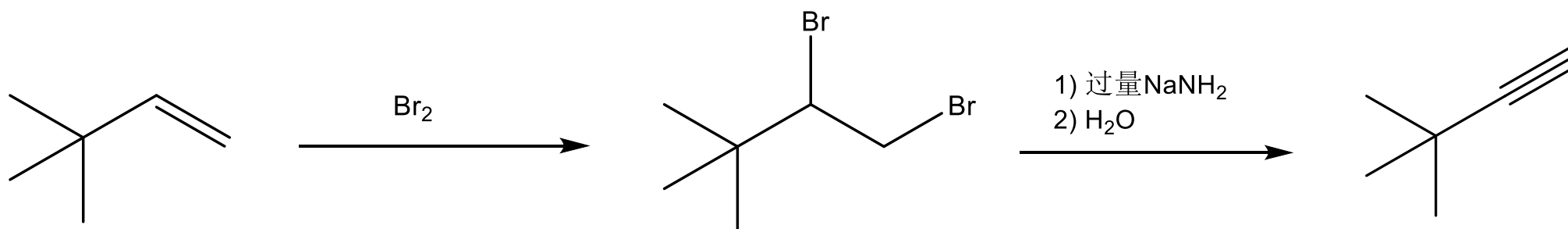
练习:



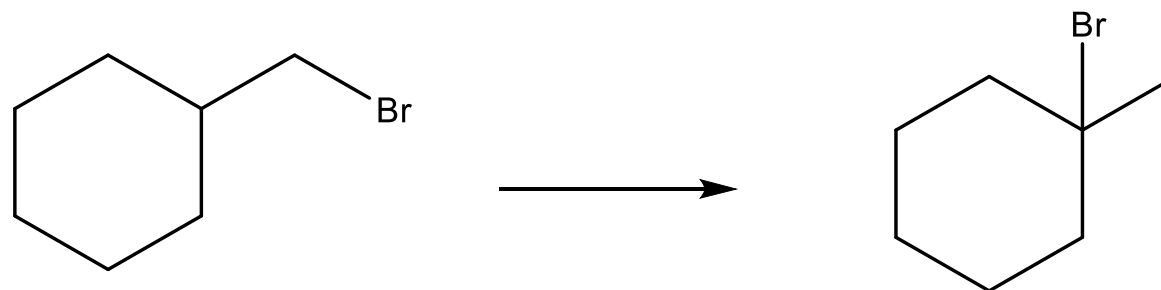
Always approach a synthesis problem by initially asking two questions:

1、 Is there a change in the carbon skeleton? No

2、 Is there a change in the functional groups? Yes



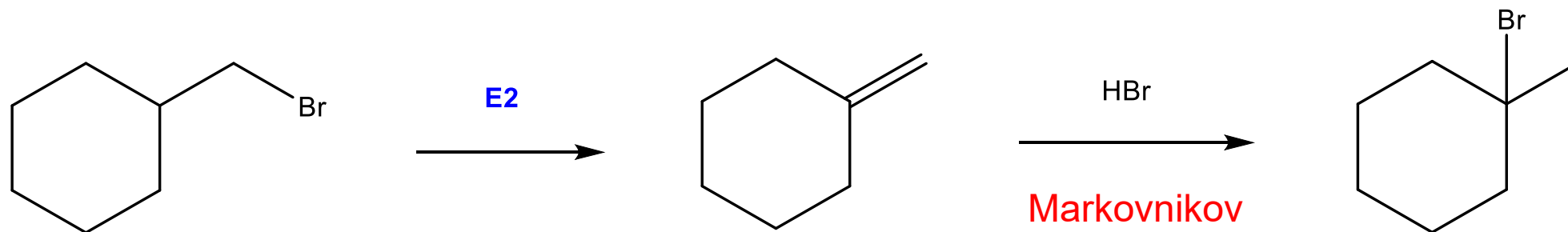
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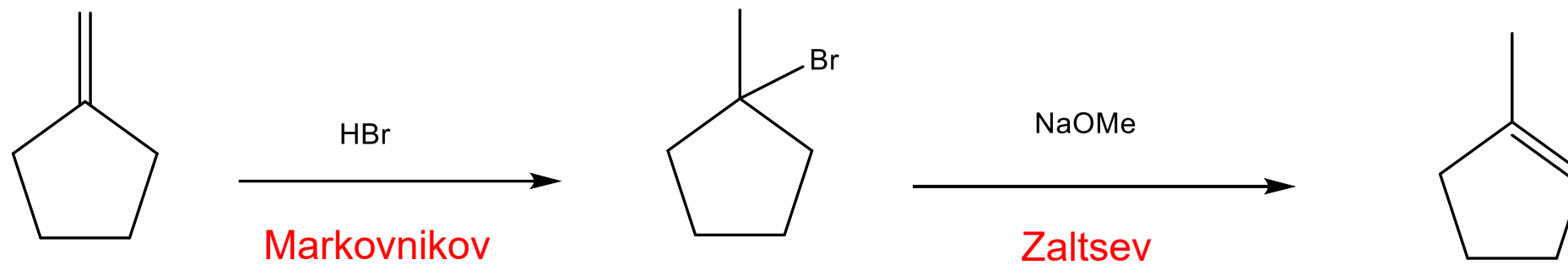
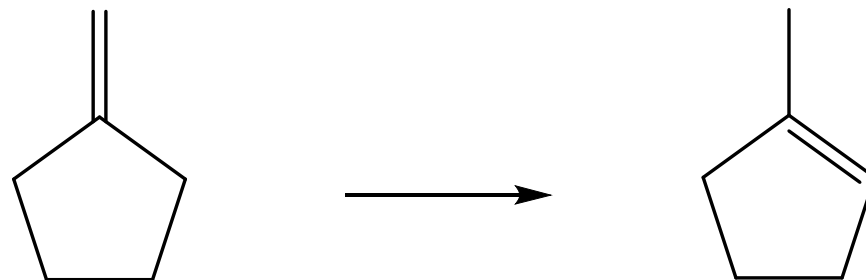
Always approach a synthesis problem by initially asking two questions:

1、 Is there a change in the carbon skeleton? No

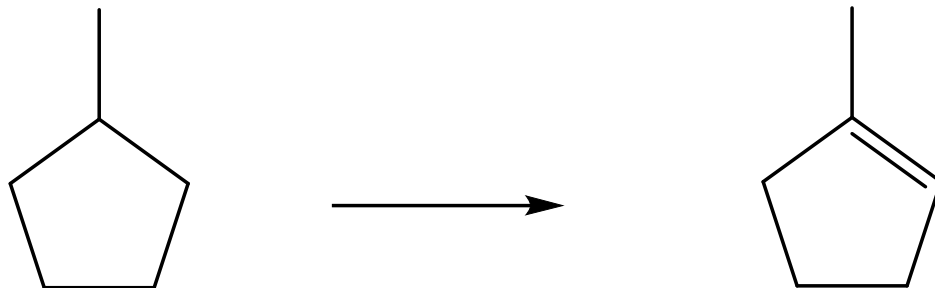
2、 Is there a change in the functional groups? Yes



练习:



练习:



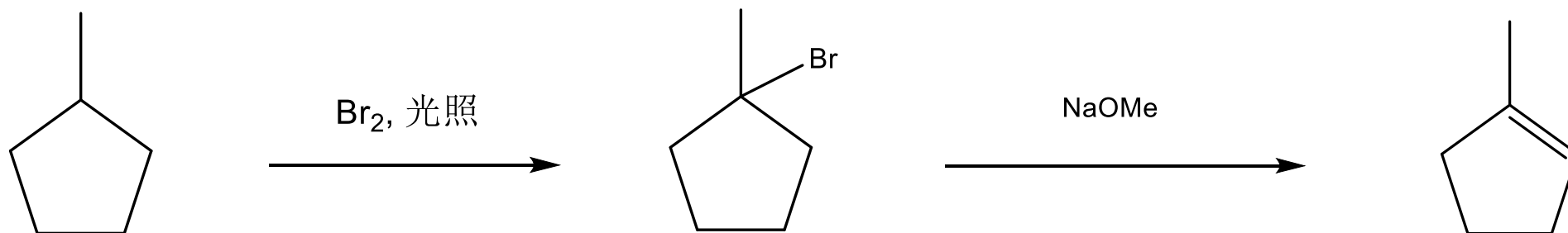
Always approach a synthesis problem by initially asking two questions:

1、 Is there a change in the carbon skeleton?

No

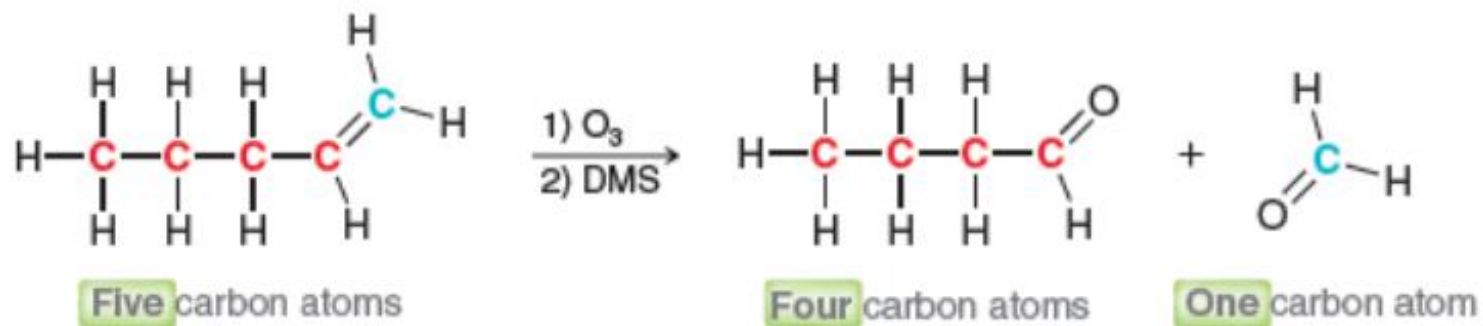
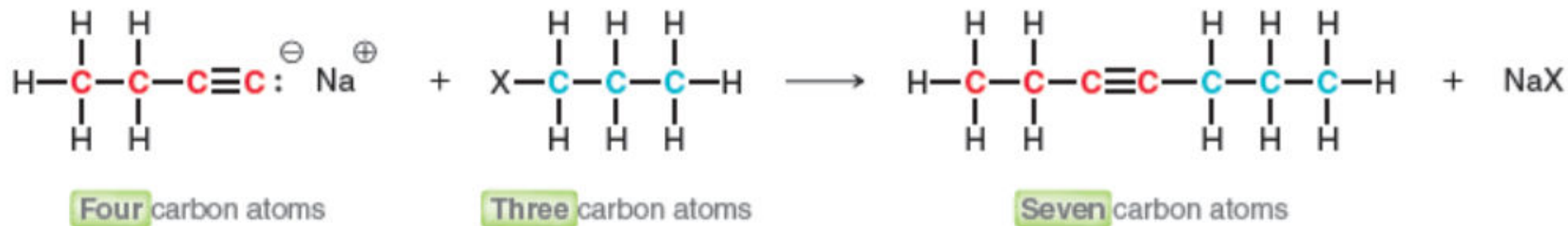
2、 Is there a change in the functional groups?

Yes, 官能团从无到有



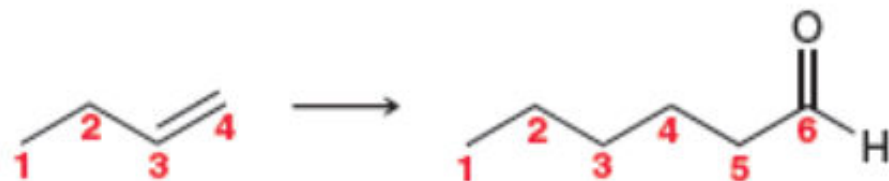
利用自由基溴代，实现了官能团的从无到有

- Reactions that change the carbon skeleton



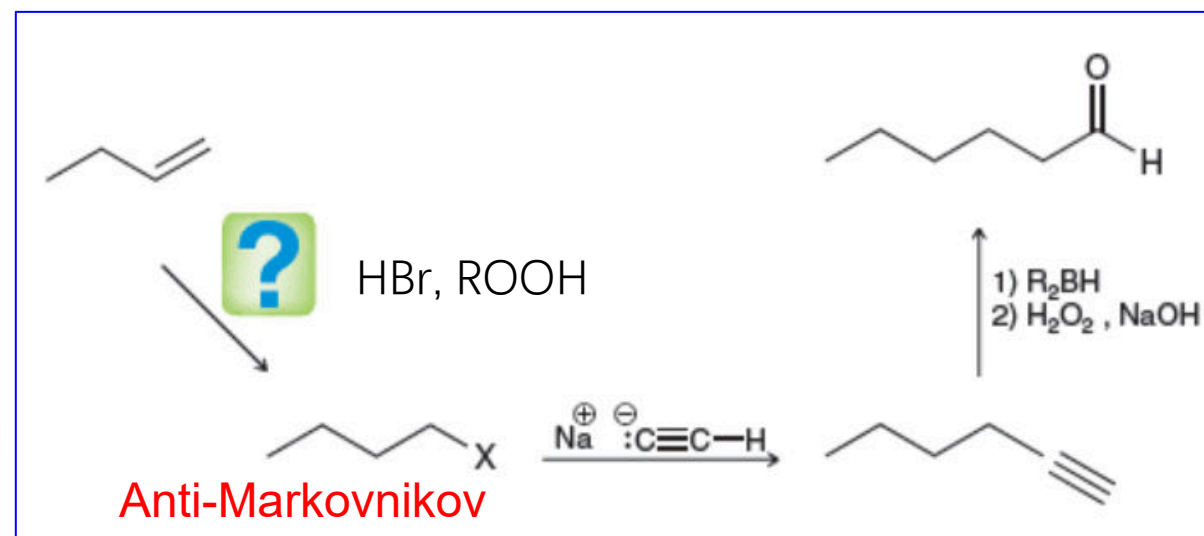
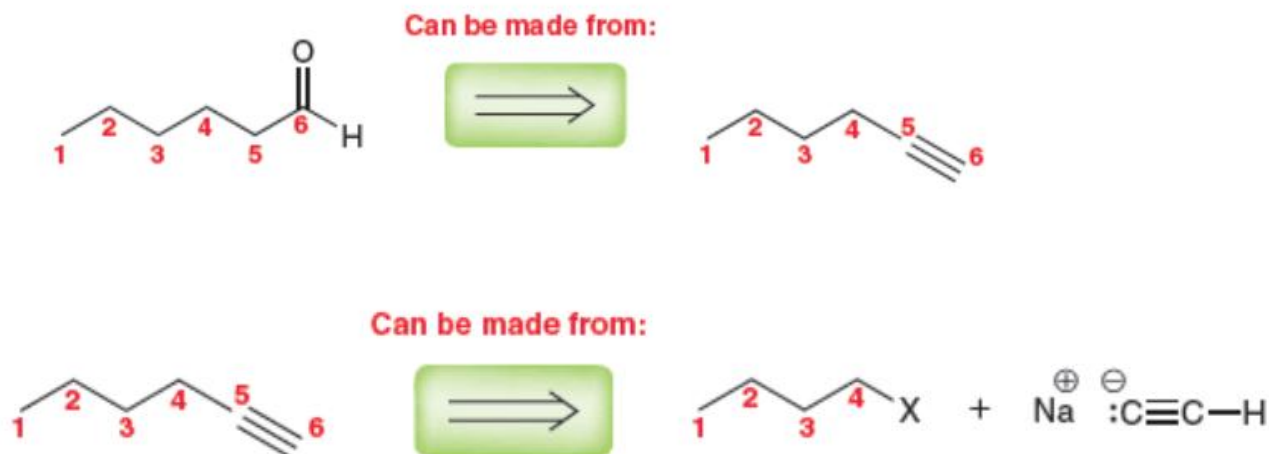
Propose a plausible synthesis for each of the following transformations:

练习:

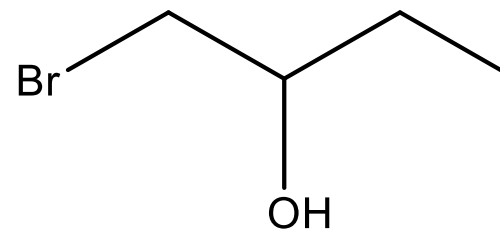
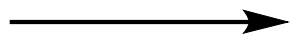


Always approach a synthesis problem by initially asking two questions:

- 1、 Is there a change in the carbon skeleton? Yes
- 2、 Is there a change in the functional groups? Yes



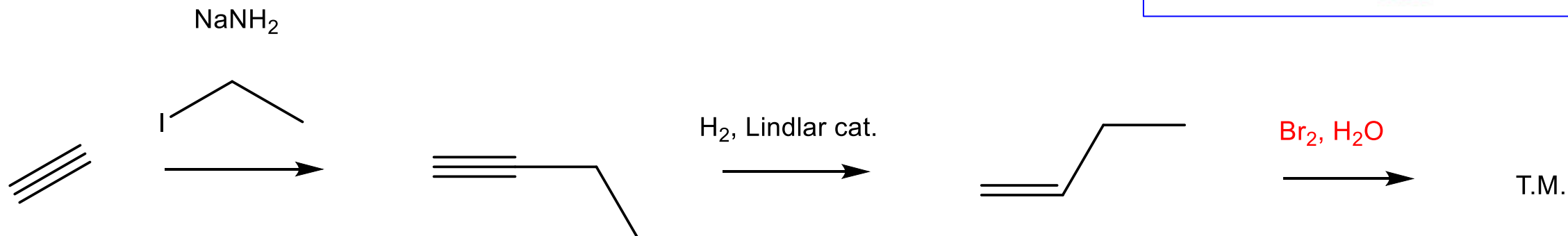
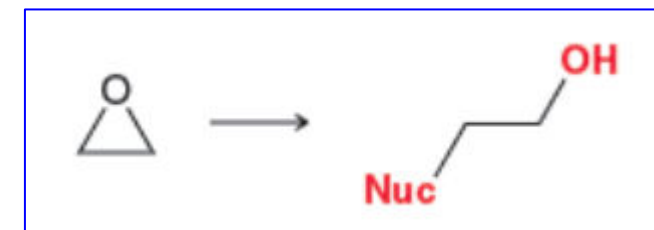
练习:



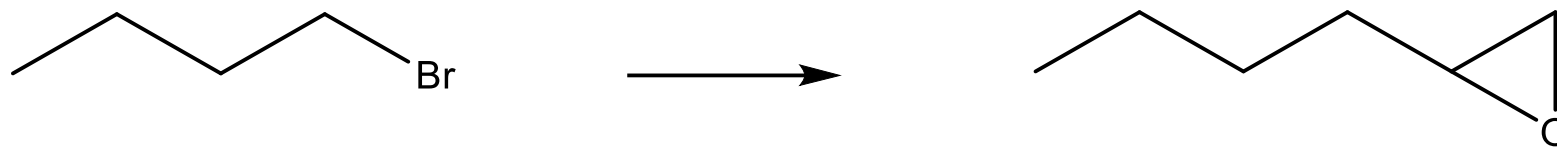
Always approach a synthesis problem by initially asking two questions:

1、 Is there a change in the carbon skeleton? **Yes**

2、 Is there a change in the functional groups? **Yes**



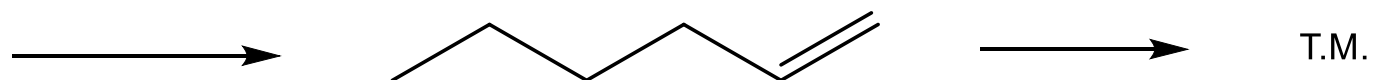
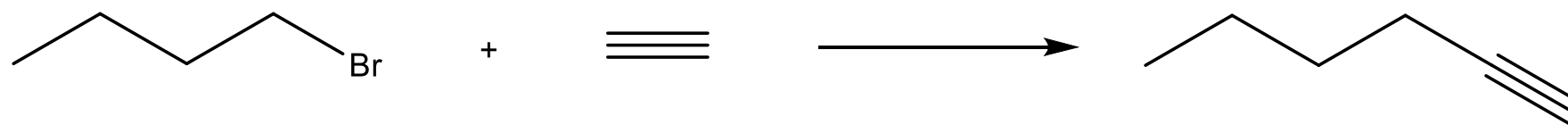
练习:



Always approach a synthesis problem by initially asking two questions:

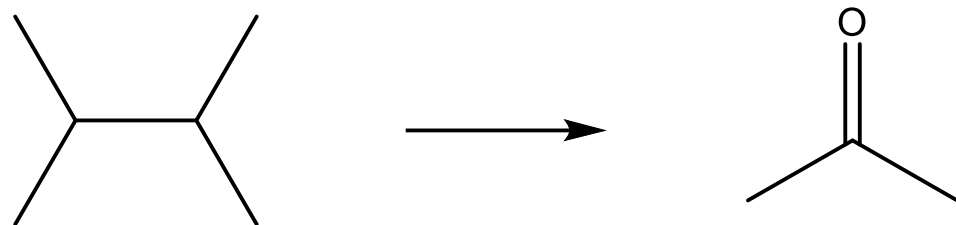
1、 Is there a change in the carbon skeleton? Yes

2、 Is there a change in the functional groups? Yes



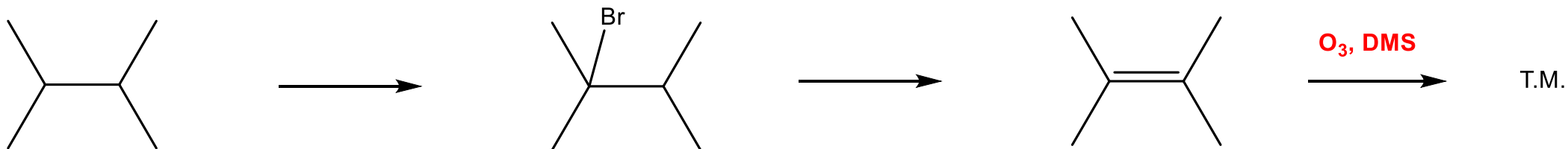
T.M.

练习:

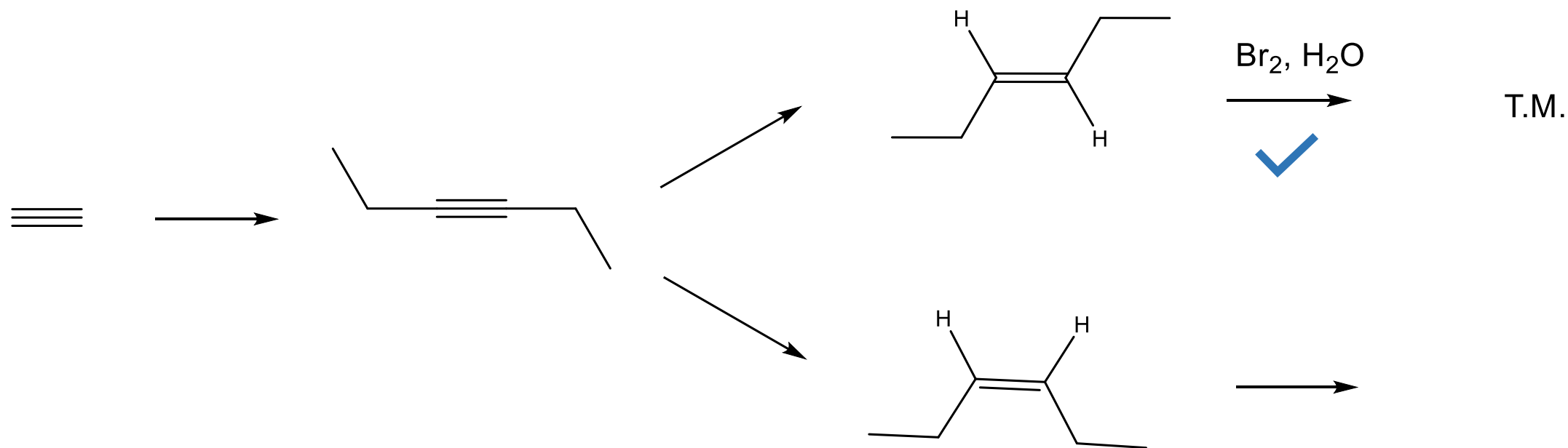
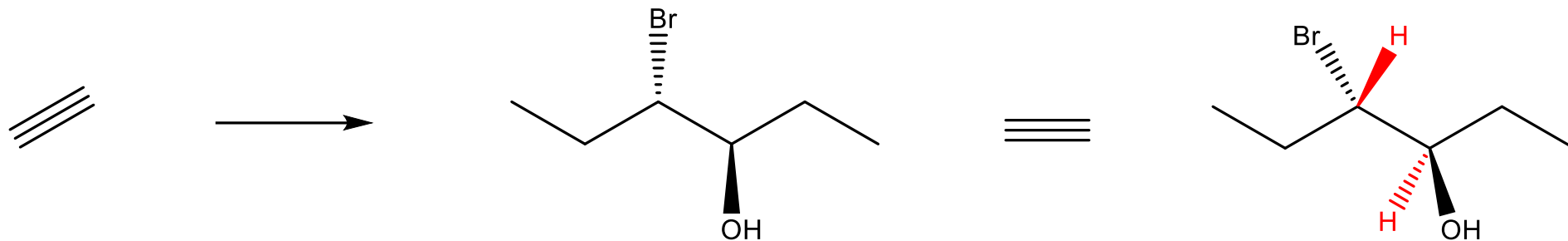


Always approach a synthesis problem by initially asking two questions:

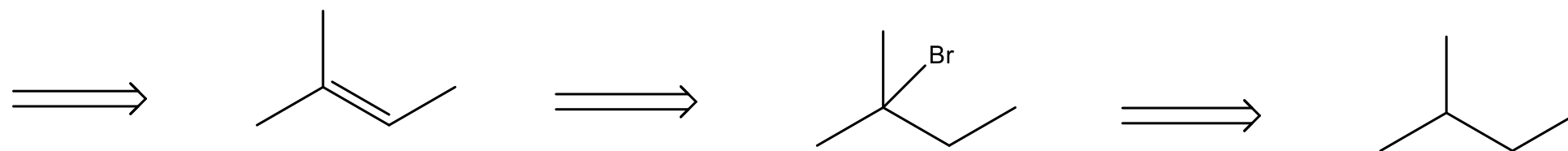
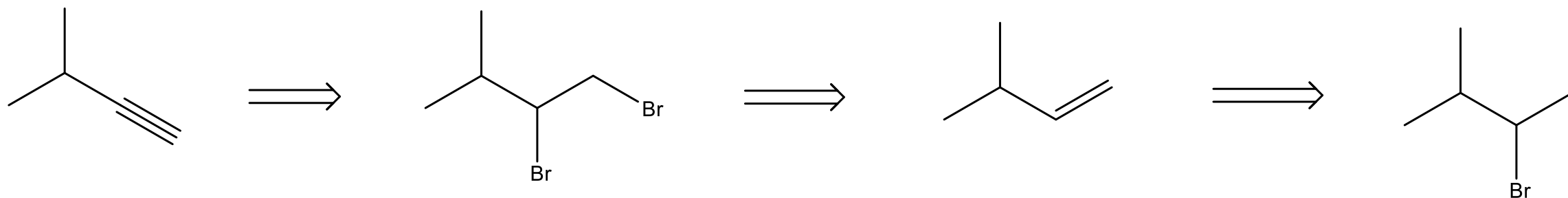
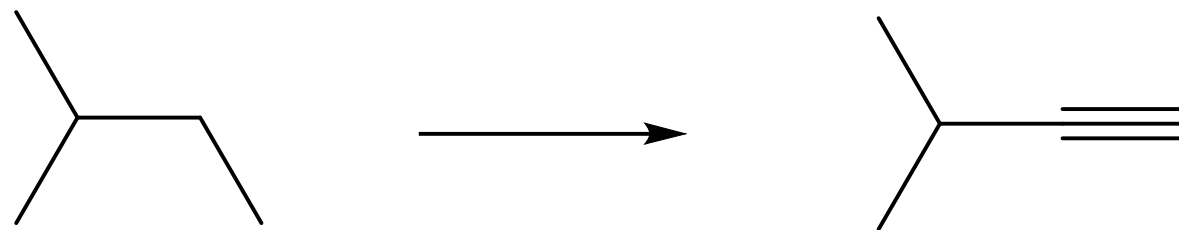
- 1、Is there a change in the carbon skeleton? Yes
- 2、Is there a change in the functional groups? Yes, 官能团从无到有

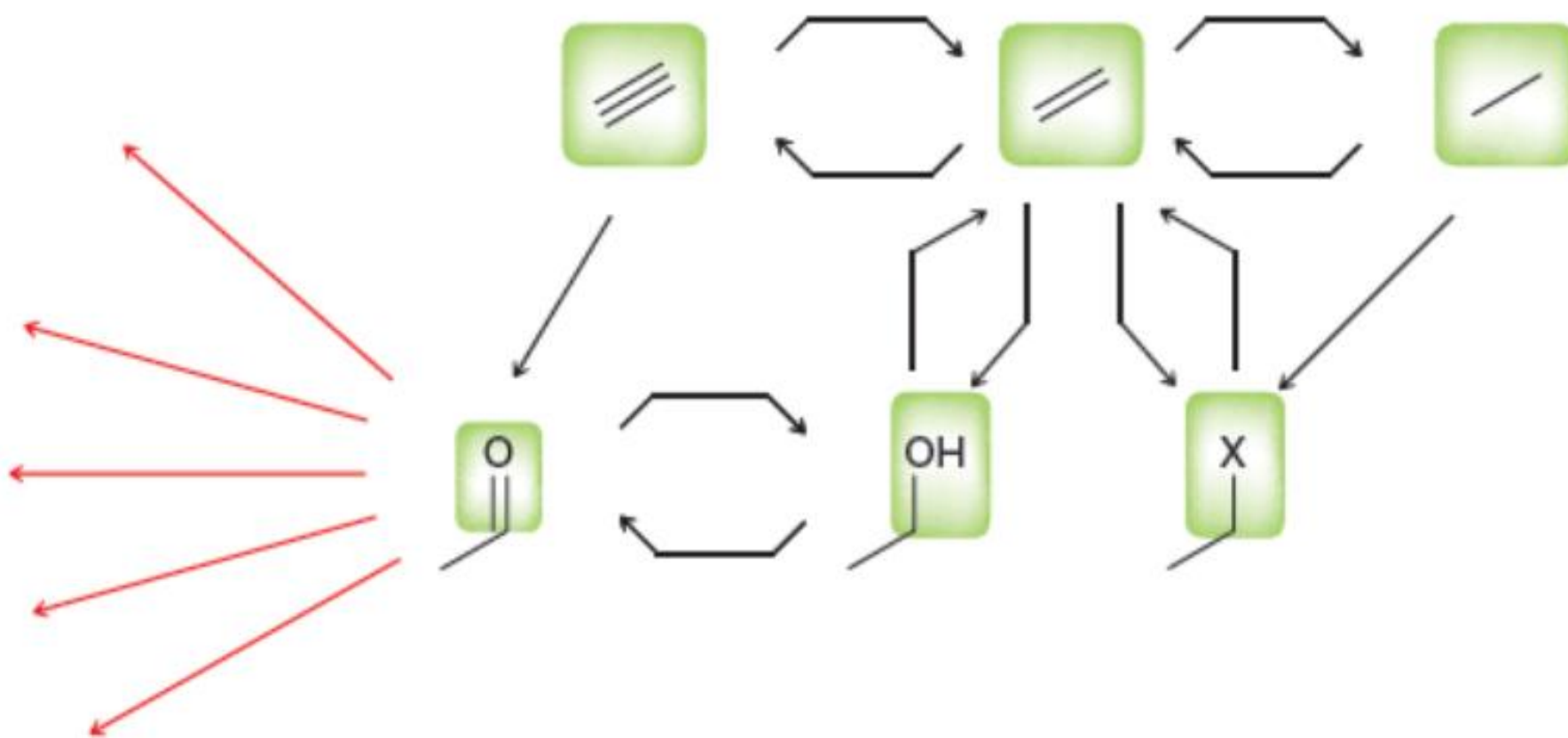


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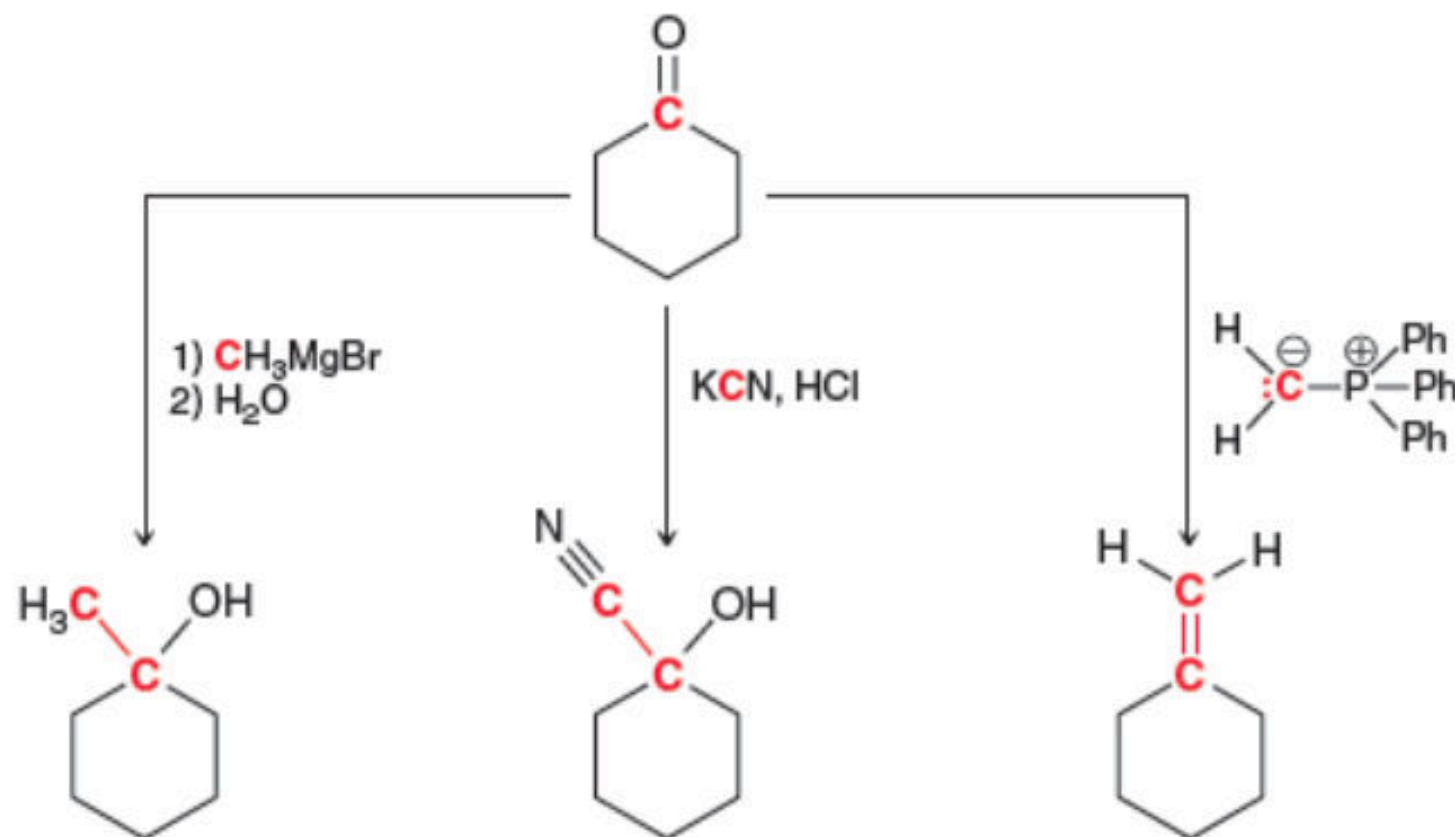


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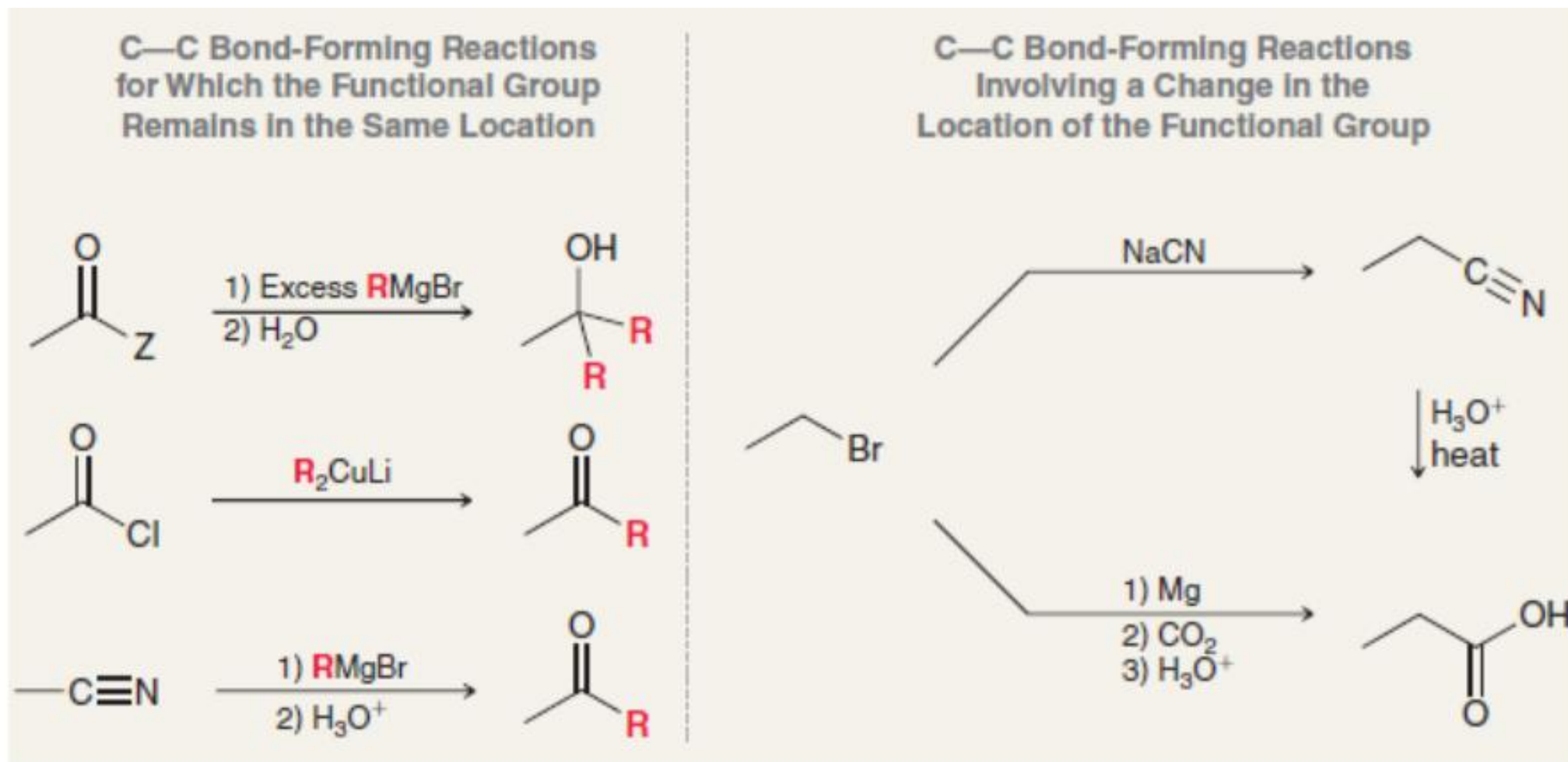




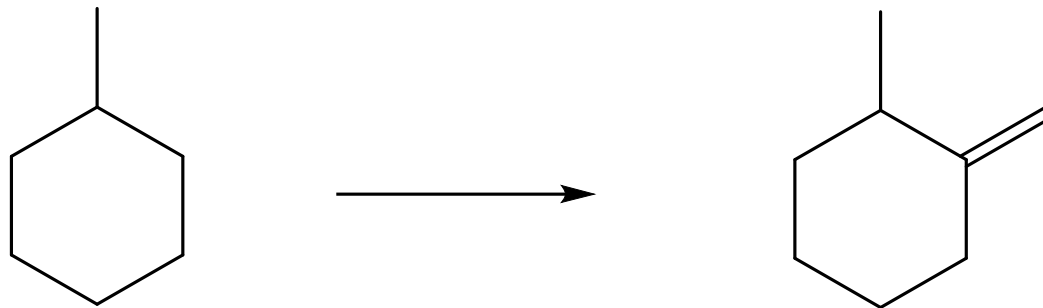
- Reactions that change the carbon skeleton



- Reactions that change the carbon skeleton



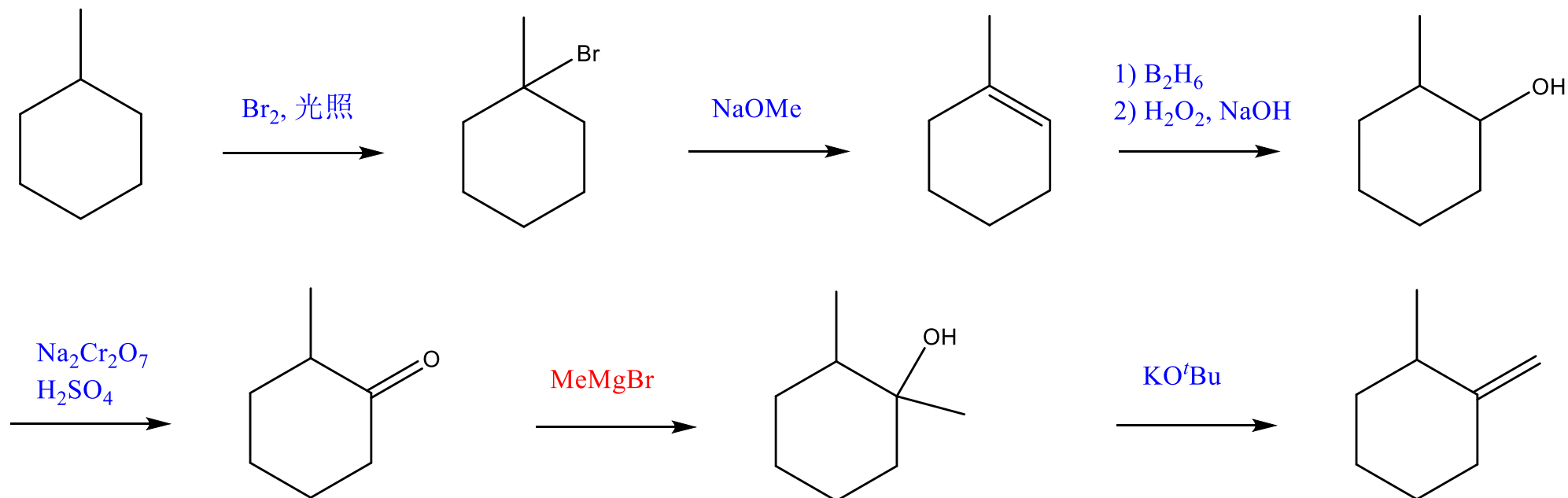
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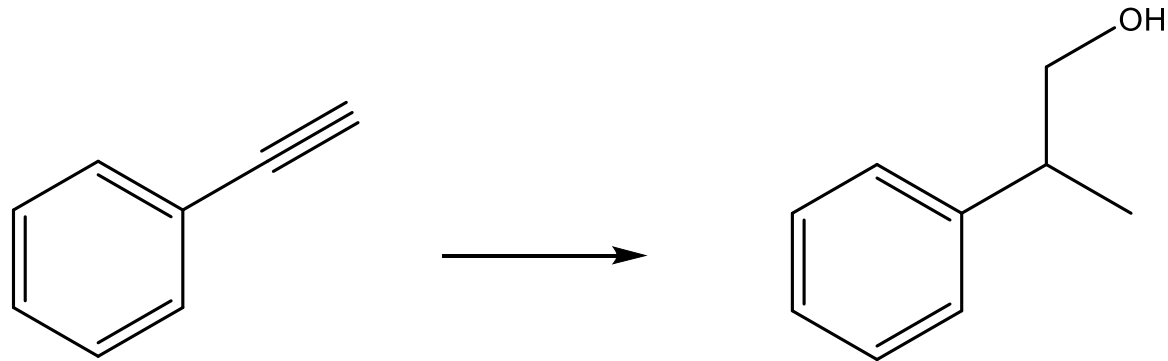
Always approach a synthesis problem by initially asking two questions:

1、 Is there a change in the carbon skeleton? **Yes**

2、 Is there a change in the functional groups? **Yes**



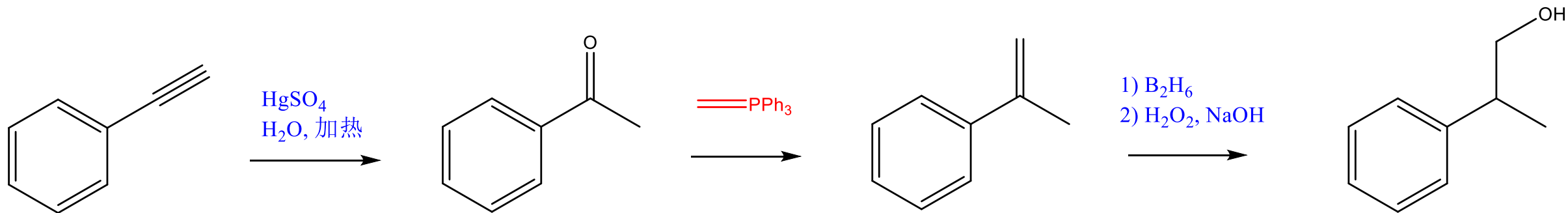
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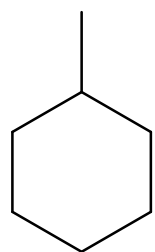
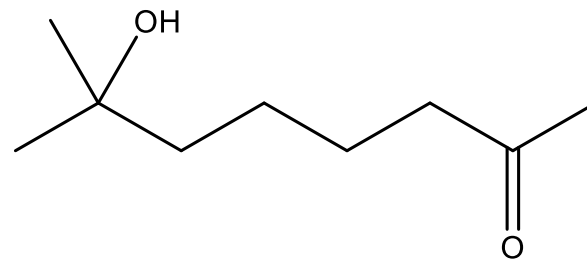
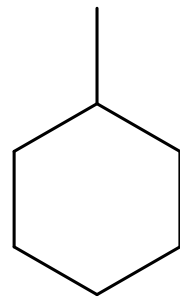
Always approach a synthesis problem by initially asking two questions:

1、 Is there a change in the carbon skeleton? Yes

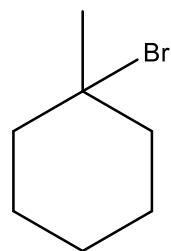
2、 Is there a change in the functional groups? Yes



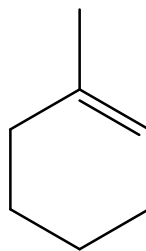
练习:



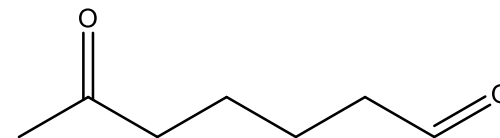
Br_2 , 光照



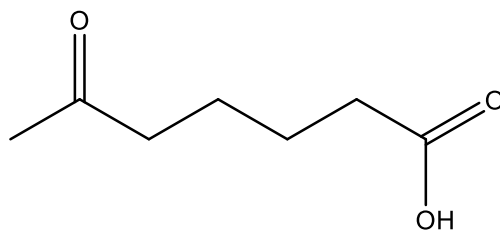
NaOMe



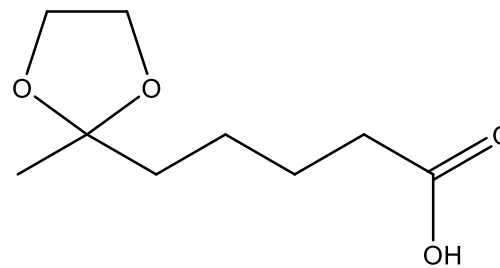
O_3 , DMS



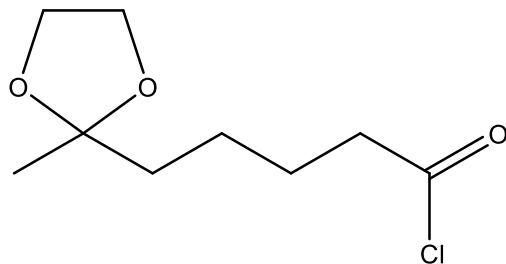
$\text{Na}_2\text{Cr}_2\text{O}_7$
 H_2SO_4



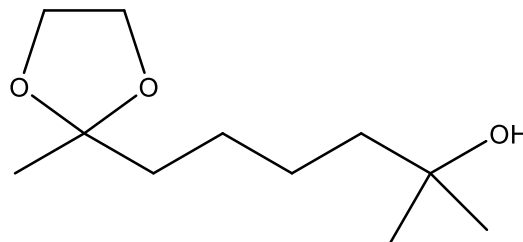
HO-CH₂-CH₂-CH₂-OH
 H_3O^+



SOCl_2

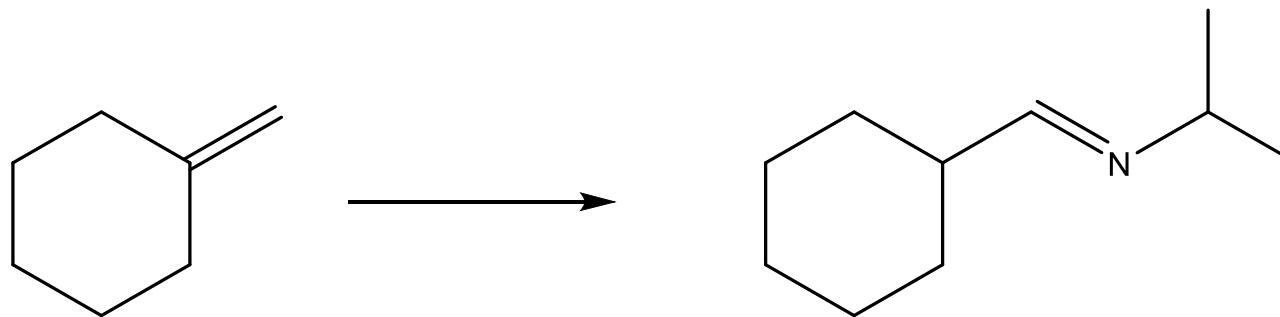


过量 MeMgBr



T.M.

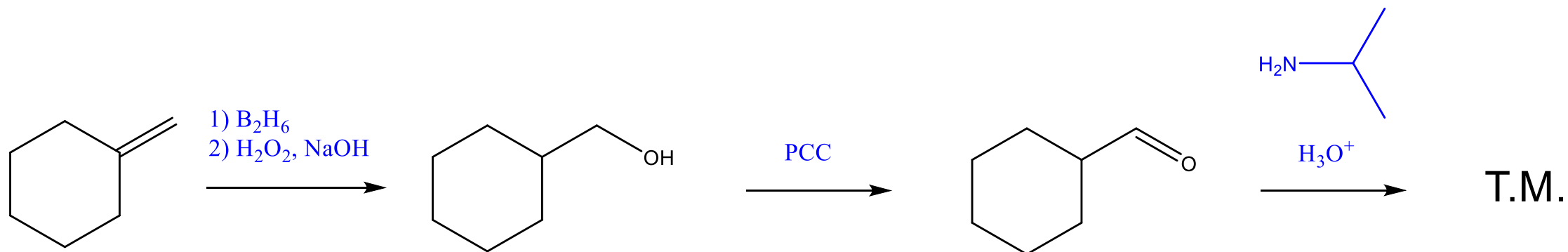
练习:



Always approach a synthesis problem by initially asking two questions:

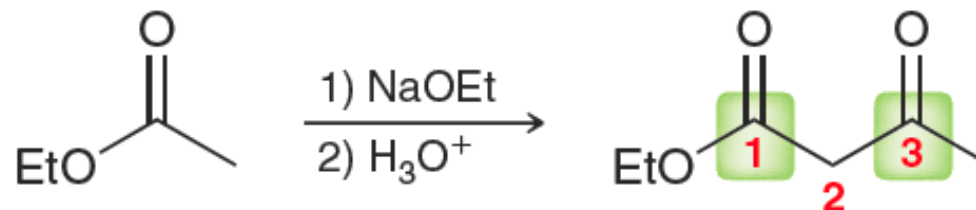
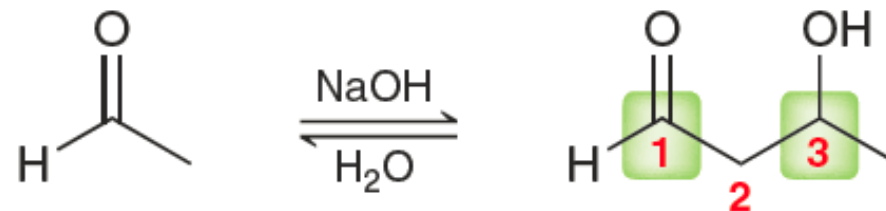
1、 Is there a change in the carbon skeleton? No

2、 Is there a change in the functional groups? Yes



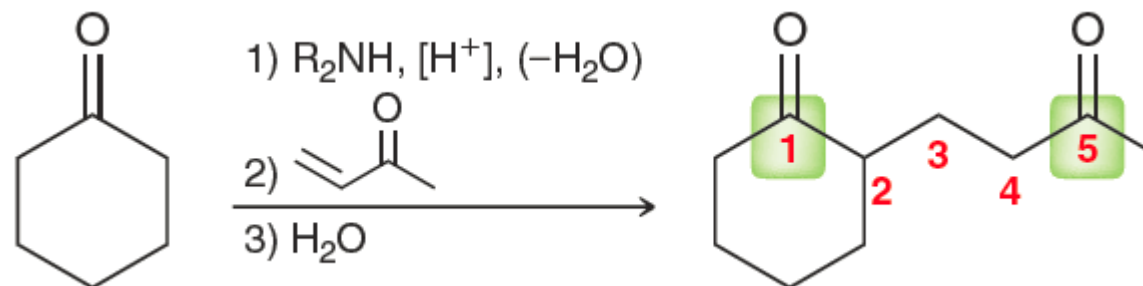
- Reactions that change the carbon skeleton

- 1, 3-双官能团化



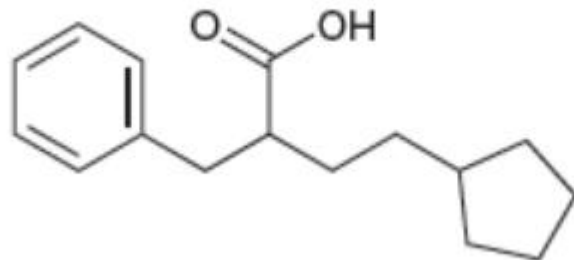
具有不同的
氧化态

- 1, 5-双官能团化

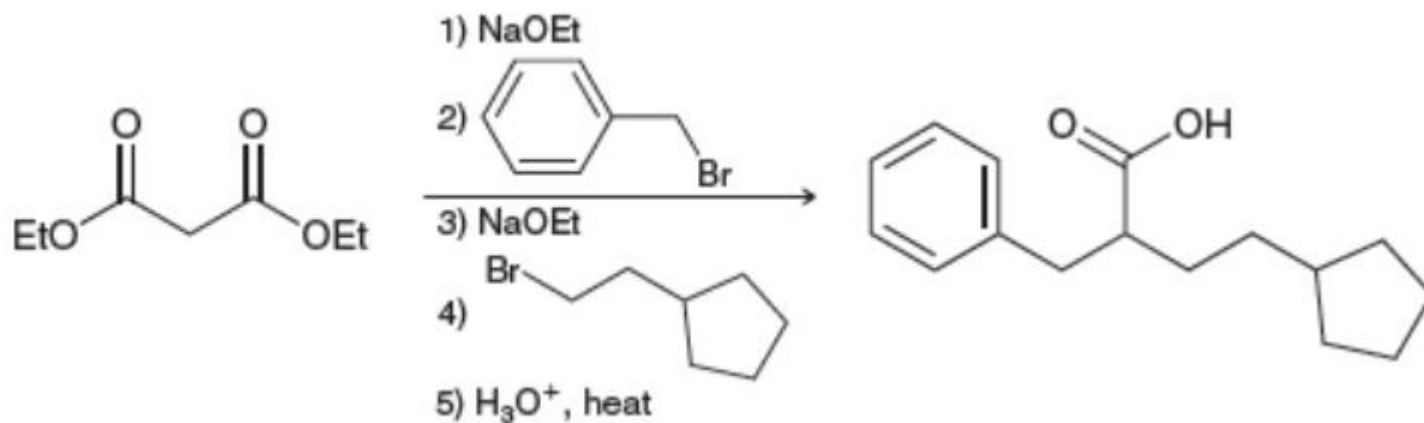
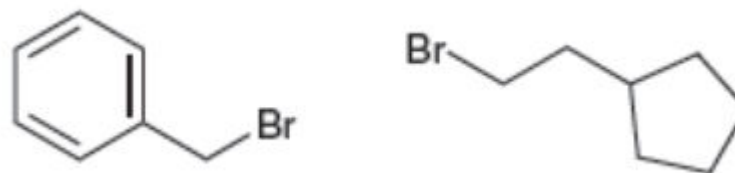
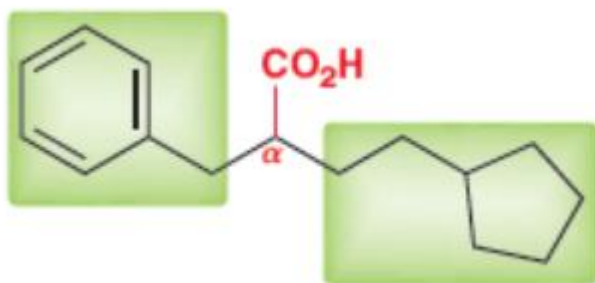


The Malonic Ester Synthesis

Show how you would use the malonic ester synthesis to prepare the following compound:

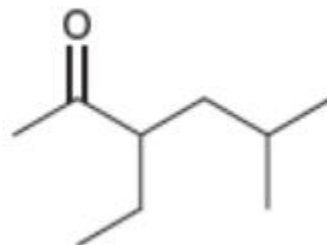


丙二酸二乙酯烷基化及水解脱羧

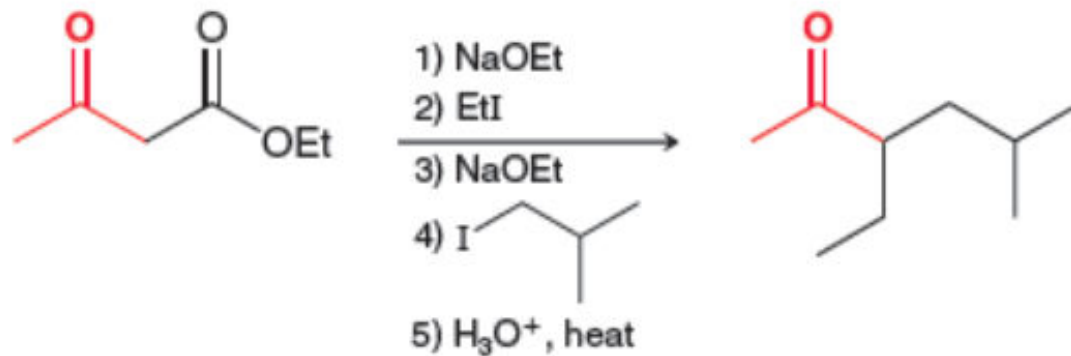
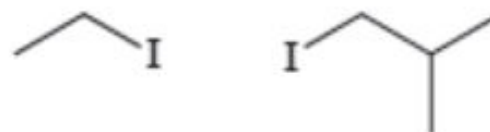
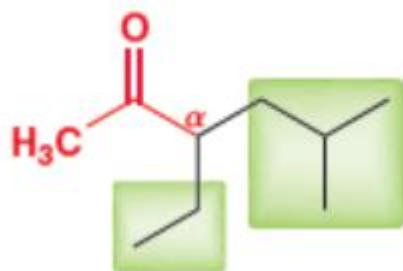


The Acetoacetic Ester Synthesis

Show how you would use the acetoacetic ester synthesis to prepare the following compound:



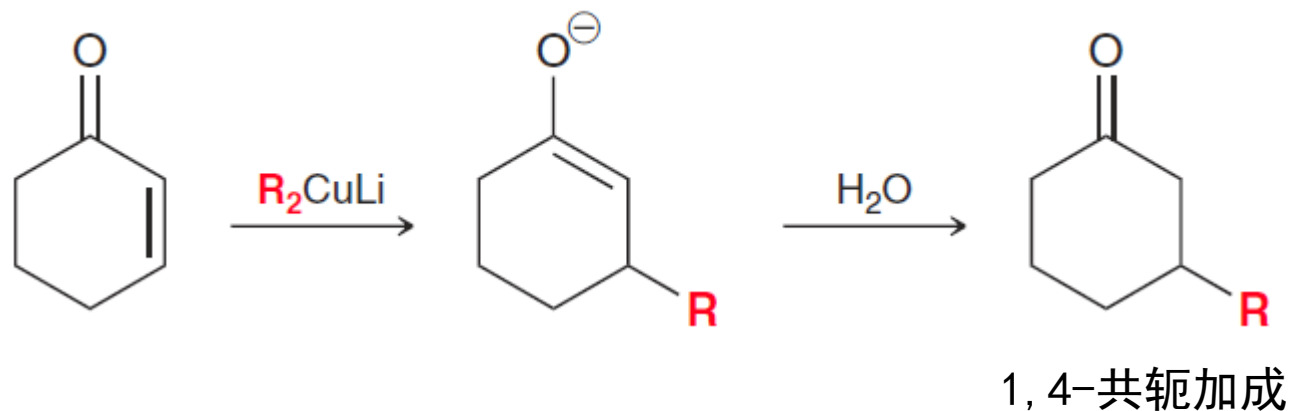
乙酰乙酸乙酯烷基化及水解脱羧



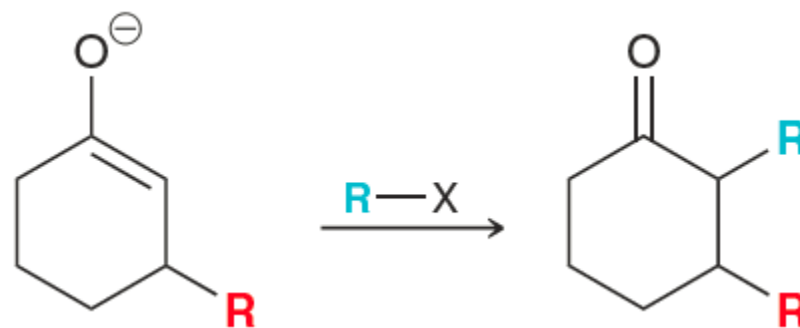
- Reactions that change the carbon skeleton

羰基化合物的烷基化

- 羰基 β -位烷基化

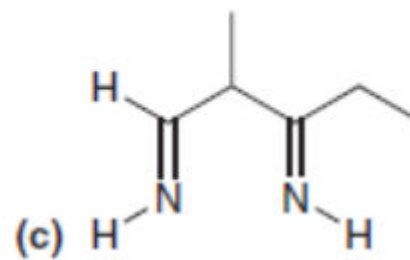
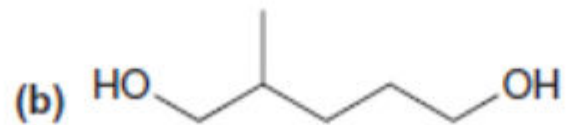
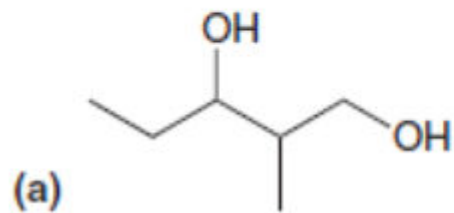


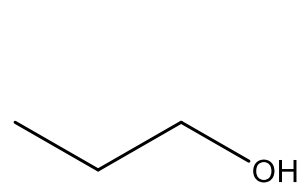
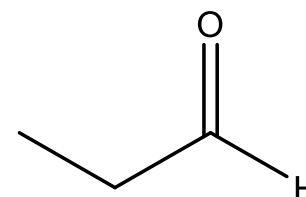
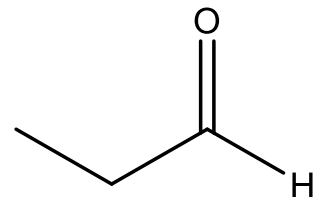
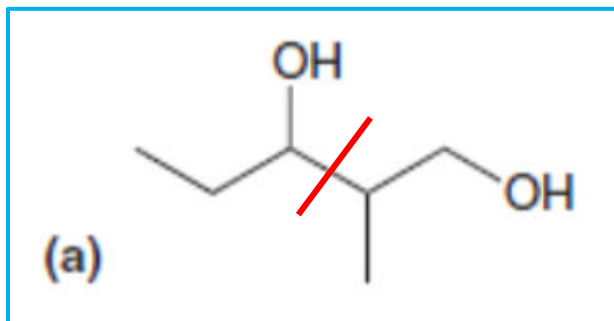
- 羰基 α 位, β -位双烷基化



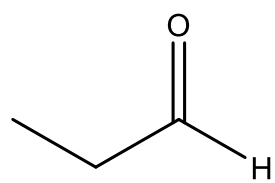
Using 1-propanol as your only source of carbon, propose an efficient synthesis for each of the following compounds:

练习:

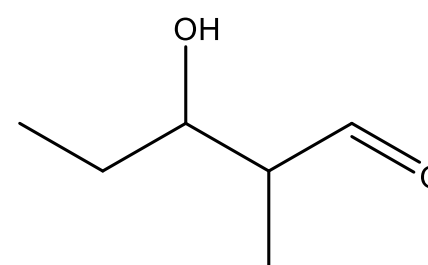




PCC

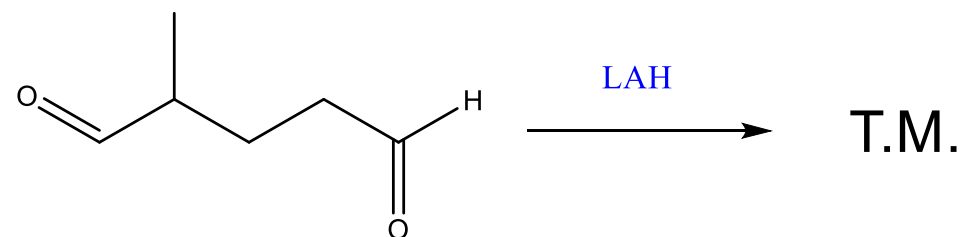
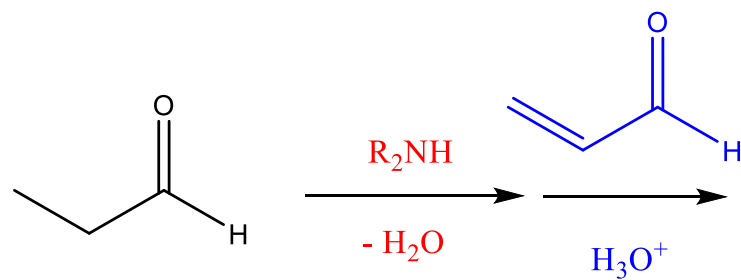
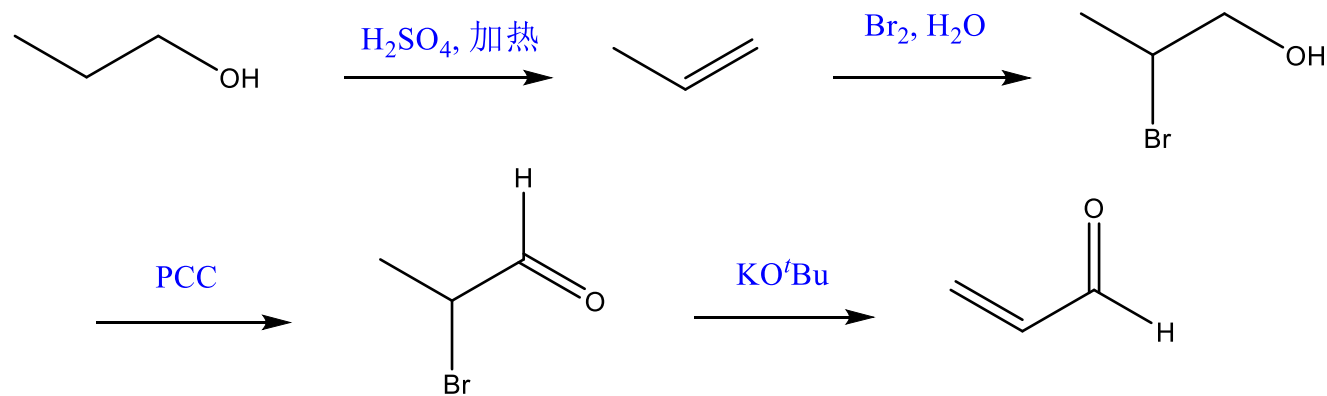
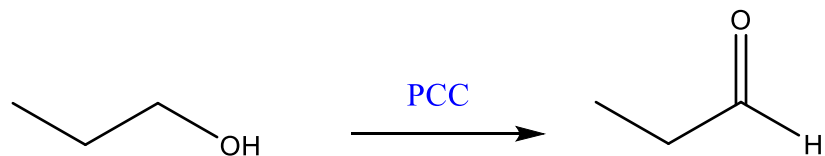
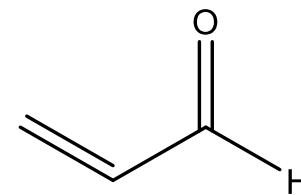
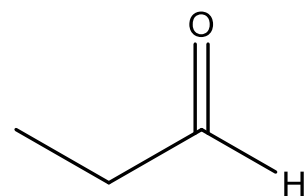
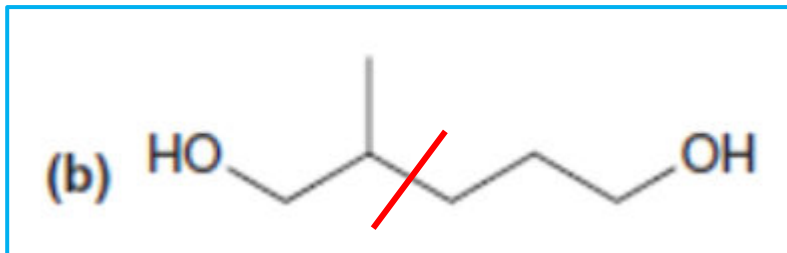


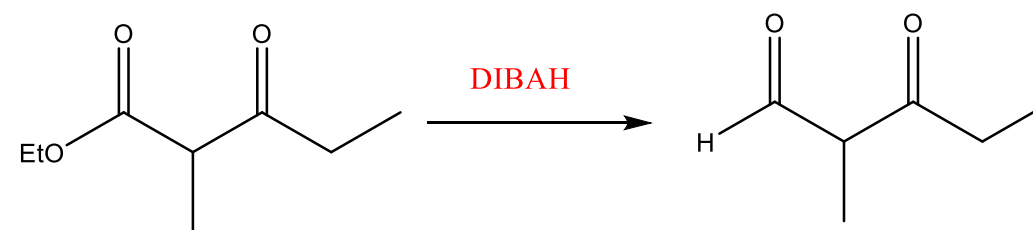
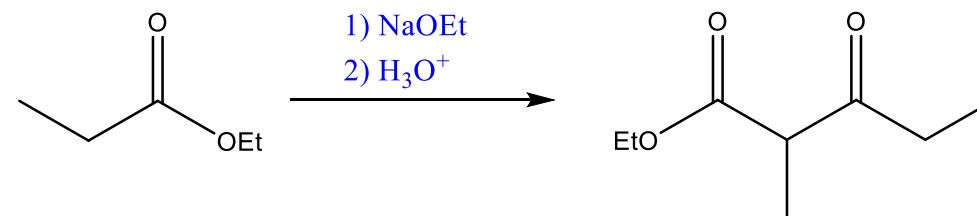
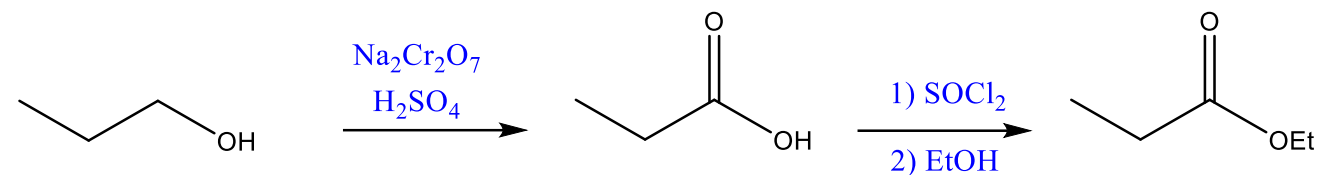
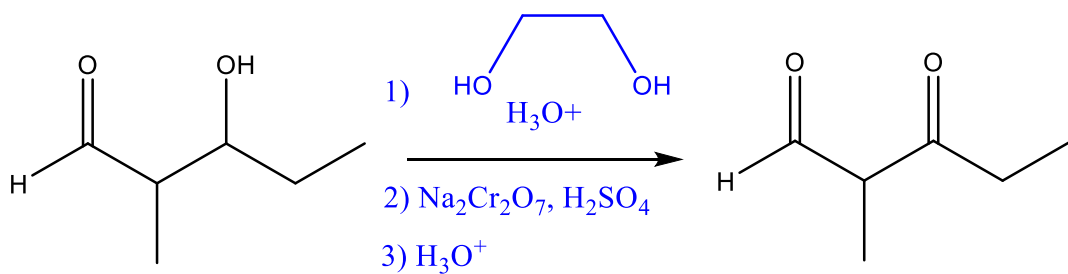
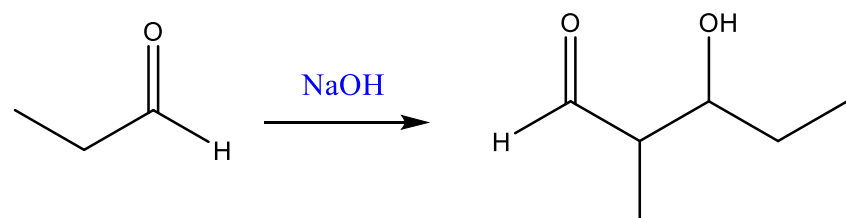
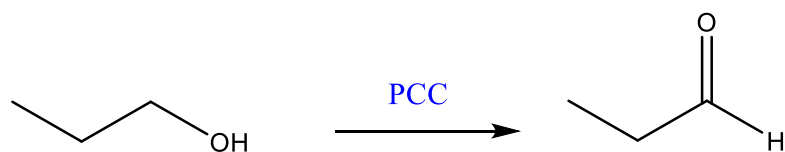
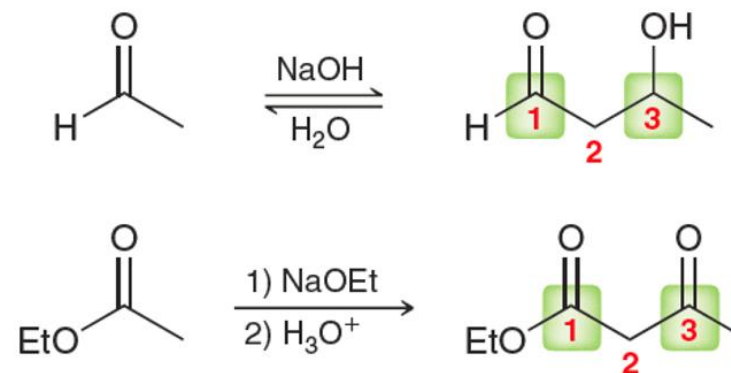
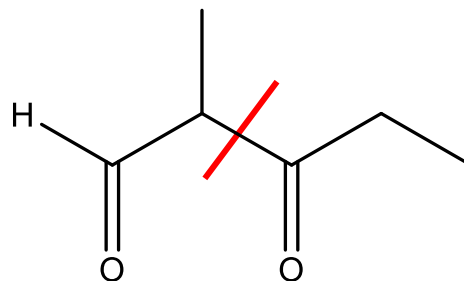
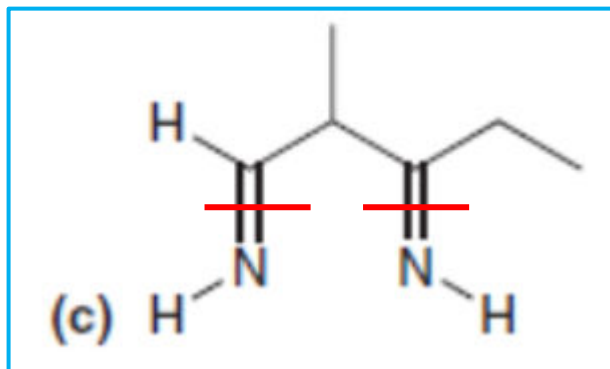
NaOH



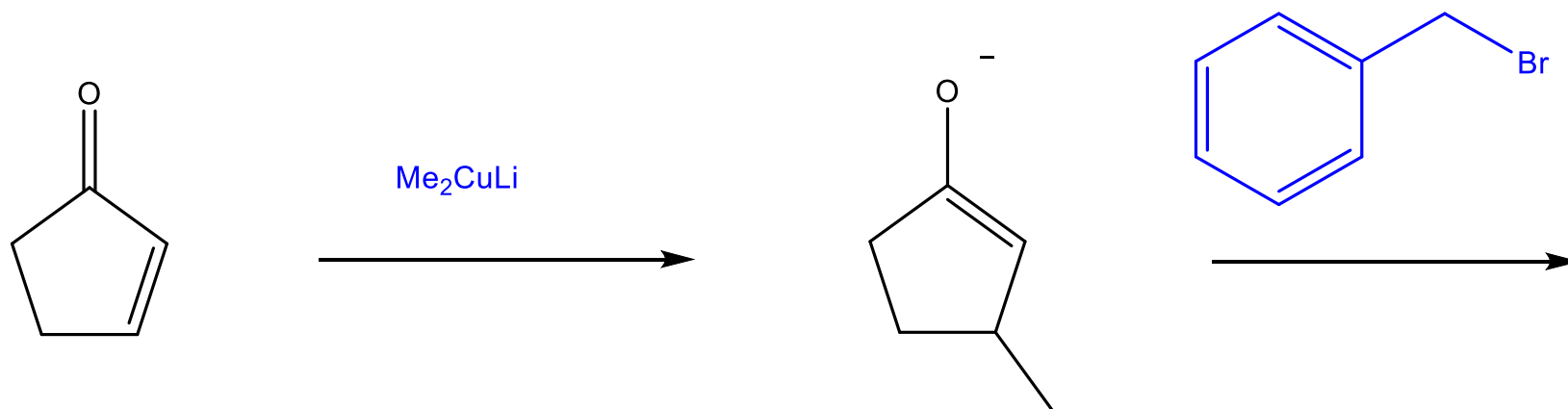
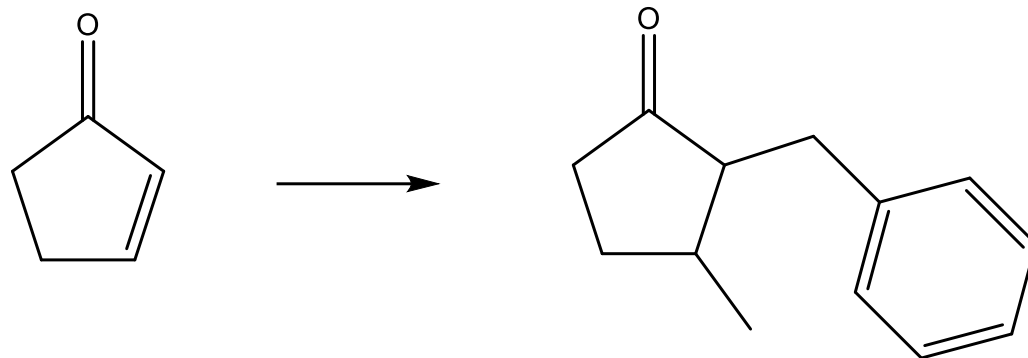
LAH

T.M.

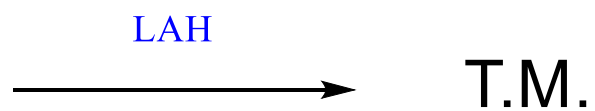
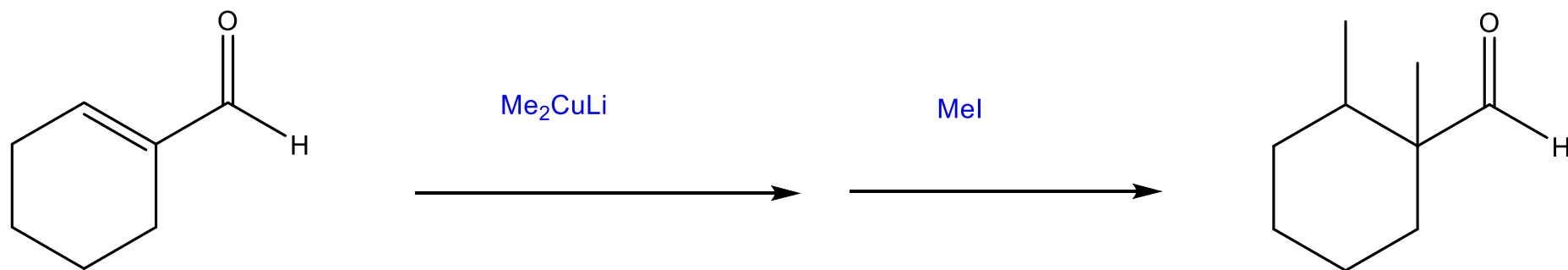
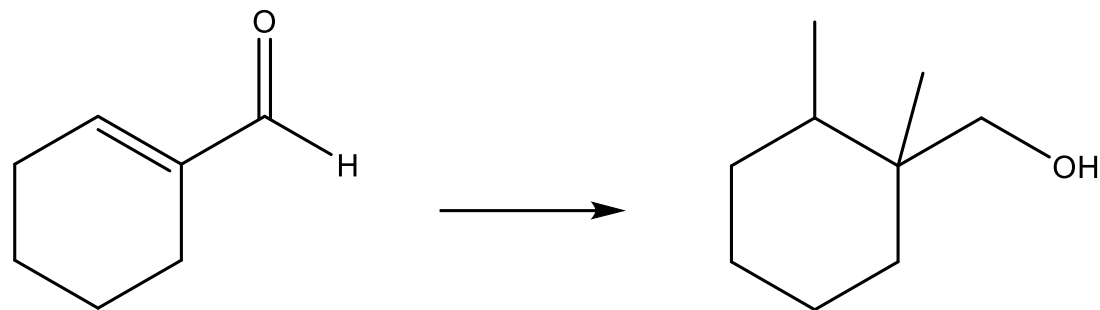


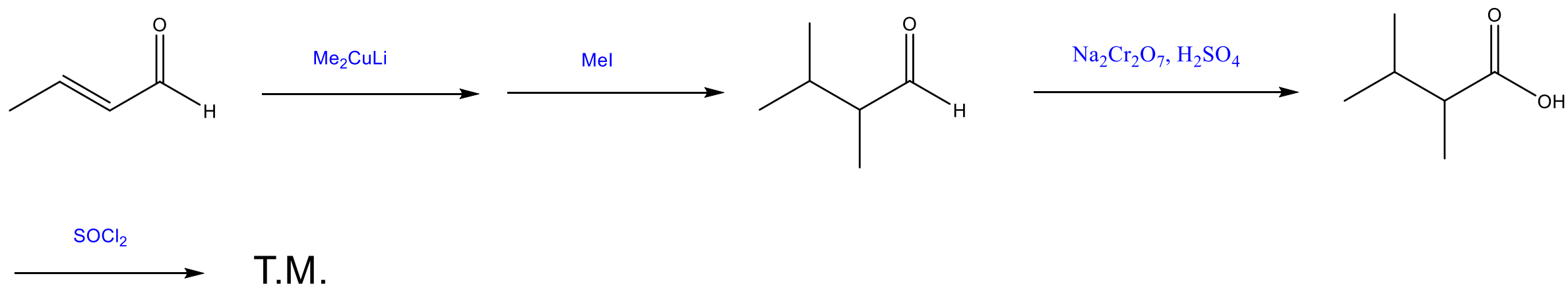
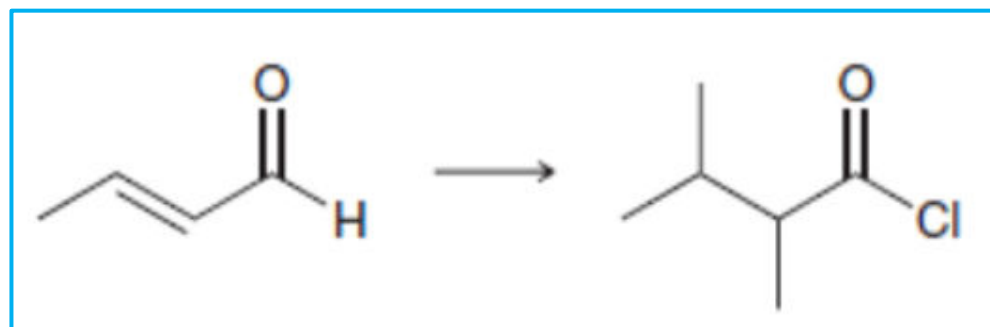


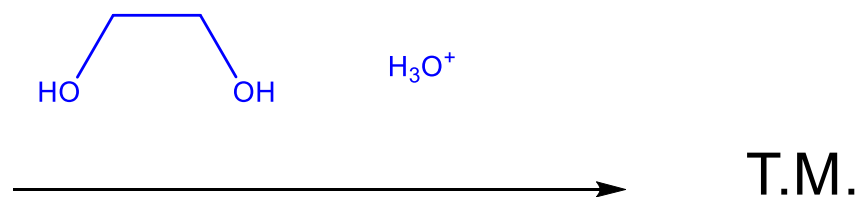
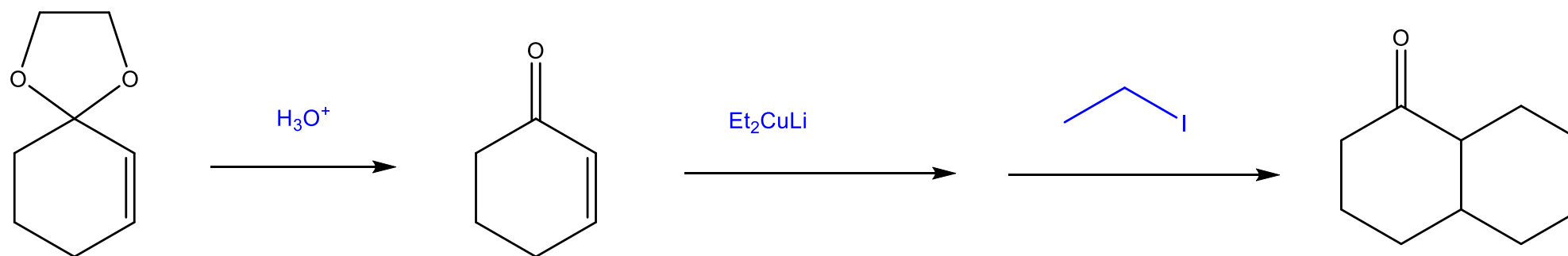
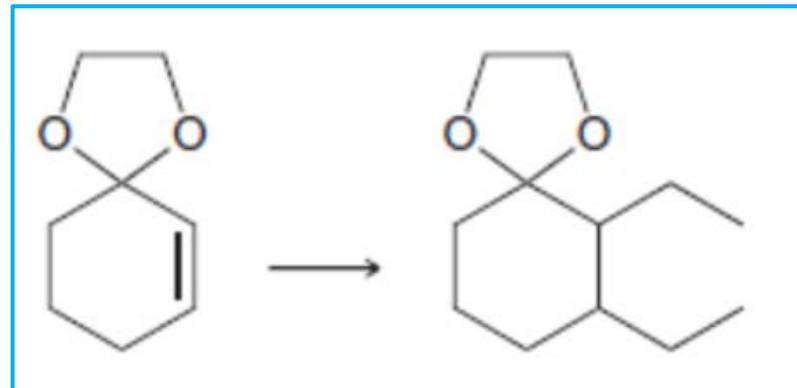
练习:



练习:







合成策略

- 基本反应（取代、加成、消除）
- 官能团位置的改变
- 碳骨架的改变
- 学会识别基本官能团模式及其构筑方法

设计合成路线很有趣，希望大家课后多多练习！

预祝大家期末考试取得好成绩！